ModelIt!

Student Activity Pack

Is Our Freshwater Running Out?

Modeling the Water Cycle, Aquifer Depletion, and Global Water Security

Grade 12L2 | NGSS HS-ESS2-5 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where does your tap water come from? Can you trace it from its source to your glass?

2 What is an aquifer and why does it matter? What happens if an aquifer is pumped dry?

3 Earth is called the water planet, but most people worry about water scarcity. How can a planet covered in water have a freshwater crisis?

4 What uses the most freshwater globally — homes, factories, or farms? How much of total freshwater do you think this use consumes?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Precipitation Rate
- Population and Agricultural Demand
- Surface Water Availability
- Groundwater Level
- Water Recycling and Efficiency
- Ecosystem Water Requirements
- Water Quality Index

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Growing Demand

Population Demand: +30% | Precipitation: Current | Efficiency: Current

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Drought Scenario

Population Demand: Current | Precipitation: -20% | Efficiency: Current

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Conservation Strategy

Population Demand: +30% | Precipitation: Current | Efficiency: +40% | Recycling: Active

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Virtual Water Trade — The hidden water embedded in traded goods — producing 1 kg of beef requires 15,000 liters of water, and water-scarce countries that import food are effectively importing the water that grew it, making global trade a major mechanism for redistributing water stress across regions

Desalination Infrastructure — The capacity to convert seawater to freshwater using reverse osmosis or thermal distillation — desalination provides a climate-independent water source but requires 3-10 kWh per cubic meter, produces concentrated brine that harms marine ecosystems, and is expensive to build and operate

Managed Aquifer Recharge — The intentional channeling of surface water into underground aquifers to accelerate natural recharge — through injection wells, spreading basins, and stormwater capture, managed recharge can partially offset extraction, but is limited by the availability of excess surface water and suitable geological conditions

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Water covers 71% of Earth\'s surface, but only 0.5% is accessible freshwater — so why are aquifers that took thousands of years to fill being drained in decades, and what happens to 2 billion people when the water runs out? Use evidence from your simulation to support your answers.

1. **Why is groundwater depletion effectively irreversible on human timescales even though the water cycle continues? What timescale mismatch does your model reveal?**

2. **Agriculture uses 70% of freshwater. What does your model suggest about the relative impact of improving irrigation efficiency versus other conservation measures?**

3. **How does climate change act as a threat multiplier for water scarcity? What two mechanisms does your model show working simultaneously?**

4. **What are the ethical dimensions of water allocation — when there isn\'t enough for everyone, who gets priority: cities, farms, industry, or ecosystems? What happens when ecosystems are sacrificed?**

5. **How does the concept of virtual water trade mean that water scarcity in one region becomes a global economic and food security issue?**

8 STEM Challenge: Design a Sustainable Water Management System

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YOUR ENGINEERING MISSION

Design a comprehensive water management system for a water-stressed region that balances human needs, agricultural demand, and ecosystem requirements while maintaining groundwater sustainability.

THE SCENARIO

A semi-arid agricultural region of 3 million people relies on a groundwater aquifer for 60% of its water supply. The aquifer level has been declining 1.5 meters per year for the past 20 years due to agricultural irrigation that far exceeds natural recharge. Climate projections indicate 15% less precipitation over the next 30 years. At current rates, the aquifer will be too depleted for economical pumping within 25 years, threatening the food production that feeds 15 million people beyond the region. Your team must design a water management system that stabilizes the aquifer while maintaining agricultural productivity and supporting population growth.

Engineering Design Thinking

1. WHAT SPECIFIC WATER CONSERVATION AND EFFICIENCY MEASURES DOES YOUR PLAN IMPLEMENT, AND WHAT PERCENTAGE OF DEMAND REDUCTION DO THEY ACHIEVE BASED ON MODEL EVIDENCE?

2. HOW DOES YOUR PLAN ADDRESS AGRICULTURAL WATER USE — THE DOMINANT DEMAND — WITHOUT DEVASTATING THE REGIONAL ECONOMY?

3. WHAT ALTERNATIVE WATER SOURCES DOES YOUR PLAN DEVELOP, AND WHAT ARE THEIR CAPACITIES, COSTS, AND ENVIRONMENTAL IMPACTS?

4. HOW DOES YOUR PLAN ALLOCATE WATER BETWEEN HUMAN CONSUMPTION, AGRICULTURE, AND ECOSYSTEM REQUIREMENTS, AND WHAT GOVERNANCE MECHANISMS ENFORCE THIS ALLOCATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Hydrologists who study the movement, distribution, and quality of water work for the USGS, state water agencies, and environmental consulting firms, earning $60,000-$120,000/year. Water resource engineers who design systems for water treatment, distribution, recycling, and desalination earn $70,000-$140,000/year. Water policy analysts who develop regulations, pricing mechanisms, and international water agreements earn $60,000-$125,000/year at government agencies and international organizations.